

What a necessity argument for $\hbar > 0$ must supply: a unit and an indeterminacy

Result, 2026-09-27; corrected the same day. Radiation assumptions (K), (W), (S), (R) define a positive universal action scale by a spectral integral (Theorem U). A separate physical principle must connect that scale to restrictions on records. Theorem I constructs arbitrarily concentrated classical posteriors when the specified pointer couplings, preparations and sequential readouts are available. A prior density ceiling alone permits this construction. A posterior ceiling alone also leaves a further task: deriving the disturbance-versus-distinguishability bound used by Theorem 6 of the Planck paper.

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: distinguish the spectral unit, posterior restrictions, and the operational disturbance bound; the three original assumptions of Theorem I need an instrument-access assumption.

The original unit/indeterminacy split and the two-pointer construction were proposed by a Fable reviewer at the user's request on 2026-09-27. The ingredients are elementary; no novelty claim is made. The new route note develops the physical candidates and the remaining operational estimate.

1. Two ingredients, and the bridge between them

The Planck paper's Theorem 6 states

$$\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \Delta(\hat{X}_j) \geq \hbar \arcsin(1 - 2\epsilon).$$

Here Δ is the standard deviation in the proof's interpolating states, with the prescribed suprema. Its proof uses both canonical translation kinematics and a bound on statistical distance generated by an observable. A conditional classical posterior variance and this operator disturbance variance are different quantities. A later estimate of a kick can reduce its conditional uncertainty; it need not erase the spread charged by that theorem.

- **(U), the unit.** Given the fixed constants of the dimensional note, $k_e/c = e^2/(4\pi\epsilon_0 c)$ has action units. Thermal radiation supplies another universal action scale, as proved below.
- **(I), the physical restriction.** A floor needs a relation between the information a record carries and its disturbance. Prior or stationary spreads provide possible ingredients; admissible subsequent observations must preserve whatever restriction is invoked.

Correction (GPT-6 Astra referee), 2026-09-27 — REJECT the universal Coulomb impulse bound: $|\Delta p|b \geq 2k_e/c$ follows only from an ideal straight-path impulse formula, not from exact scattering.

For a prescribed path (ut, b) and coupling $k = |n_1 n_2| k_e$, integration of the transverse Coulomb force gives

$$|\Delta p_\perp| = \int_{-\infty}^{\infty} \frac{k b dt}{(b^2 + u^2 t^2)^{3/2}} = \frac{2k}{bu}.$$

This is the small-deflection approximation for scattering, or an exact force integral along an externally maintained path. Charge atomicity and $u < c$ give the proposed inequality within that prescription. For exact repulsive Rutherford scattering with reduced mass μ , $\tan(\Theta/2) = k/(\mu u^2 b)$, so the magnitude of the full momentum change satisfies

$$b|\Delta p| = \frac{2k/u}{\sqrt{1 + [k/(\mu u^2 b)]^2}} \longrightarrow 0 \quad (b \downarrow 0).$$

An infinitely heavy repulsive centre with $u \ll c$ supplies this counterexample within non-relativistic mechanics: the particle slows near the centre. Finite particle sizes would add a model-dependent cutoff. In either calculation a known trajectory gives a calculable impulse; its magnitude supplies no universal stochastic spread.

2. Theorem U: a normalized unit from radiation thermodynamics

Let $u(\nu, T) d\nu$ be a nonnegative measurable equilibrium energy density at $T > 0$. Assume:

- **(K)** the same spectrum for all material enclosures (Kirchhoff universality);
- **(W)** $u(\nu, T) = \nu^3 f(\nu/T)$ (Wien scaling; Wien 1893, reprint, metadata);
- **(S)** $0 < \int_0^\infty u(\nu, T) d\nu < \infty$;
- **(R)** $u(\nu, T)/[(8\pi\nu^2/c^3)k_B T] \rightarrow 1$ as $\nu/T \rightarrow 0$, with fixed $c, k_B > 0$.

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: replace the dimensional heuristic by an integral definition; a finite nonnegative spectrum and a ratio asymptotic are explicit, and the wavelength-peak formula needs a peak hypothesis.

Theorem U. Define $x = \nu/T$ and

$$r(x) = \frac{c^3 x f(x)}{8\pi k_B}, \quad I = \int_0^\infty x^2 r(x) dx, \quad \beta = I^{-1/3}, \quad h_{\text{rad}} = k_B \beta.$$

Then $0 < I < \infty$, $[\beta] = \text{K s}$, and $h_{\text{rad}} > 0$ is a universal action scale independent of the enclosure. With $\varphi(y) = r(y/\beta)$,

$$u(\nu, T) = \frac{8\pi k_B T \nu^2}{c^3} \varphi(\beta \nu/T), \quad \lim_{y \downarrow 0} \varphi(y) = 1, \quad \int_0^\infty y^2 \varphi(y) dy = 1.$$

Writing $\int u d\nu = \sigma' T^4$ gives

$$h_{\text{rad}} = \left(\frac{8\pi k_B^4}{c^3 \sigma'} \right)^{1/3}.$$

Proof. Substitution $\nu = Tx$ gives $\int u d\nu = (8\pi k_B/c^3) T^4 I$. Assumptions (S) and nonnegativity make I finite, and (R), $r(x) \rightarrow 1$, makes it positive. Universality of f makes I enclosure-independent. Its units are $(\text{K s})^{-3}$. Changing variable $y = \beta x$ proves the normalized shape and the formula for h_{rad} . $\square$

This proof uses measurability and positivity inherent in an energy spectrum, and the constants explicitly present in (R). It makes no assumption that a physical law is built from a finite list of dimensional monomials. Kirchhoff universality allows dependence on other universal constants, including charge or particle masses; it establishes enclosure independence rather than independence from every microscopic constant. The normalization is conventional: any fixed positive numerical multiple of h_{rad} is an equally universal action unit.

If $y^4 \varphi(y)$ has a specified finite positive maximizer y_* , then the spectrum per unit wavelength peaks at $\lambda_{\text{max}} T = b$, where

$$h_{\text{rad}} = \frac{y_* k_B b}{c}.$$

Integrability and (R) alone permit narrow high-frequency spikes and need not give such a maximum or a unique one. For the Planck shape in its usual normalization $\varphi_P(y) = y/(e^y - 1)$, the integral is $\pi^4/15$, so $h_P = (\pi^4/15)^{1/3} h_{\text{rad}}$ and its wavelength peak has $y_{*,P} \simeq 4.965$. The measured spectral shape fixes this numerical conversion. The original Planck citation here was to the earlier Wien-law entropy work, Ann. Phys. 306, 69; the Planck distribution is reported in Planck 1901, Ann. Phys. 309, 553–563 (metadata). No quantization of a body follows from choosing this unit.

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: neutral probes alone do not remove charge dependence, and an exponential radiation tail alone does not derive corpuscle quantization.

With both h_{rad} and k_e/c available, a proposed floor can depend on their dimensionless ratio. Neutral composites may retain charged constituents and polarizability. To claim a pure-number multiple shared by all records requires an additional universality principle. Likewise, reading the action unit as a per-corpuscle law $p = h_P/\lambda$ needs physical assumptions beyond (K)–(R). The interval of fits and corpuscle momenta in Newton’s optics remain separate source quantities.

3. Theorem I: the two-pointer construction

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: ACCEPT the support-width algebra; specify the available Hamiltonians, freely squeezable preparations, and retained first reading.

Theorem I. Suppose a classical instrument class permits (i) Hamiltonian dynamics of body and pointers, (ii) the product rectangular preparations below, each canonical pair having density at most $1/X$, (iii) finite-resolution coordinate readouts interpreted by Bayesian conditioning, and (iv) the two specified impulsive couplings and storage of the first reading before the second coupling. There are no additional restrictions on squeezing, apparatus energy, gains or readout resolution. Then for every $X > 0$ and $\eta > 0$ it contains a protocol whose final body posterior has support widths $\Delta q \Delta p < \eta$ on every record of positive probability.

Proof. Put the body uniformly on $[0, a] \times [0, X/a]$, the first pointer (y, p_y) on $[0, \epsilon] \times [0, X/\epsilon]$, and a second independent pointer (z, p_z) on $[0, \epsilon_2] \times [0, X/\epsilon_2]$. The first Hamiltonian pulse, integrated strength $g > 0$, is gqp_y . Its canonical map is $y \mapsto y + gq$, $p \mapsto p - gp_y$, with q, p_y fixed. Store a reading of y in an interval of width ϵ' . It confines q to width $(\epsilon + \epsilon')/g$.

The second pulse $g_2 p_y p_z$, $g_2 > 0$, sends $z \mapsto z + g_2 p_y$ and $y \mapsto y + g_2 p_z$. It leaves q, p, p_y fixed; the stored earlier reading is retained. A reading of z in a bin of width ϵ'_2 confines p_y to width $(\epsilon_2 + \epsilon'_2)/g_2$. Thus the body’s final momentum $p_{\text{initial}} - gp_y$ has width at most $X/a + g(\epsilon_2 + \epsilon'_2)/g_2$, and

$$\Delta q \Delta p \leq \frac{X(\epsilon + \epsilon')}{ag} + \frac{(\epsilon + \epsilon')(\epsilon_2 + \epsilon'_2)}{g_2}.$$

Choosing small positive ϵ, ϵ' at fixed finite values of the other parameters makes this less than η . Both pulses preserve phase-space volume. All preparation densities meet the ceiling; the unread conjugate ranges increase as their coordinate ranges decrease. $\square$

A posterior supported in such a rectangle cannot obey a fixed density ceiling $1/X$ when its area is less than X . This is a counterexample to preservation of a *prior-only* ceiling by the specified instruments. Hamiltonian dynamics as an abstract property alone does not guarantee access to these pulses. A physical theory can instead restrict its admissible instruments, their composition, or the recording memory.

4. What a record-floor principle must restrict

Correction (GPT-6 Astra referee), 2026-09-27 — REJECT the exhaustive three-way list and the automatic inference of Theorem 6: instrument availability is a fourth premise, and posterior concentration and disturbance cost require distinct bounds.

A universal posterior restriction excludes at least one premise of the *four-part* Theorem I. A theory can keep classical Bayesian inference and product preparations while constraining readouts and their composition. Its disturbance bound then needs an operational proof.

- *Quantum kinematics* changes the classical phase-space state and transformation structure: there is generally no positive joint distribution for all canonical observables with

unrestricted classical conditioning. Bayesian updating of ordinary classical records remains valid. Saying every read gives a kick supplies no quantitative lower bound on an unknown kick.

- *A random background* can correlate stationary bodies and pointers. The presence of an unobserved background alone neither forbids all product preparations nor stops inference of its effects through a body. Marshall 1963 (publisher abstract through Crossref) starts with oscillator distributions matched to quantum ground states and derives a sustaining field. Boyer 1975 (publisher abstract) supplies a Lorentz-invariant random boundary field and sets its scale by $\hbar$. These source claims leave an independent posterior ceiling and its constant to be proved.
- *An epistemic restriction* can constrain every preparation, transformation and measurement. Bartlett, Rudolph and Spekkens 2012 (author abstract) impose a covariance uncertainty condition and maximum entropy and obtain Gaussian quantum statistics. This specifies much more than a pointwise density ceiling. Matching its freely chosen scale to h_{rad} remains a physical hypothesis.
- *Newton's optics* provides determinate ray mechanisms and measured fringe geometry. Even a deterministic bending law with nonzero recoil is compatible with classical conditioning. Its bearing on an irreducible unknown disturbance is tested in the new route note.

5. Consequence for STATE

Theorem U supplies an enclosure-independent radiation action scale with an explicit normalization. Theorem I identifies a concrete set of classical instruments that defeats a prior-only ceiling. The next necessity step must justify a restriction on admissible records and the associated disturbance-versus-distinguishability bound. The new route note evaluates the background and inflexion candidates and states the missing estimate precisely.